



Marine Biology

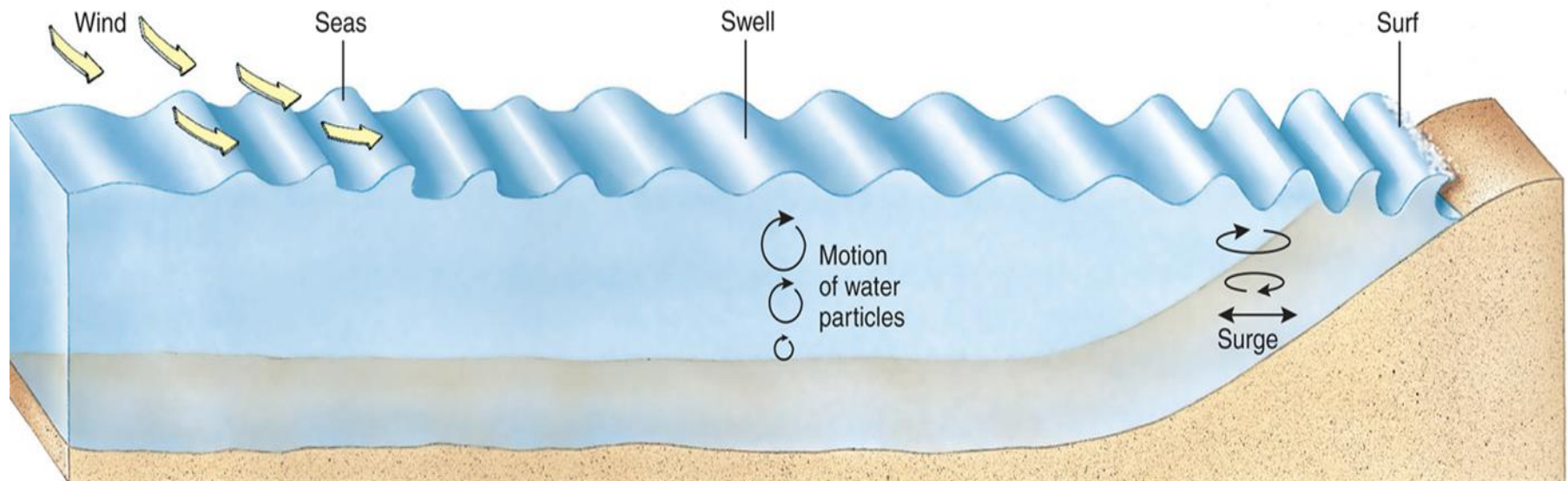
海洋生物学



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- **Waves** are the result of wind blowing over the water's surface. The size of waves depends on how long and fast the wind blows; longer and faster equals a larger wave
- ❑ **Swell** (涌浪) is a long series of ocean waves, generally produced by wind, and lasting after the wind has ceased
- ❑ As waves near the shore (shallower water), the bottom of the wave “drags” the bottom, this forces the waves to slow and move closer together (shorter wavelength, **surge**, 近岸浪)
- ❑ Eventually, the “drag” causes the wave crest to fall to form wave break/**surf** (拍岸浪)

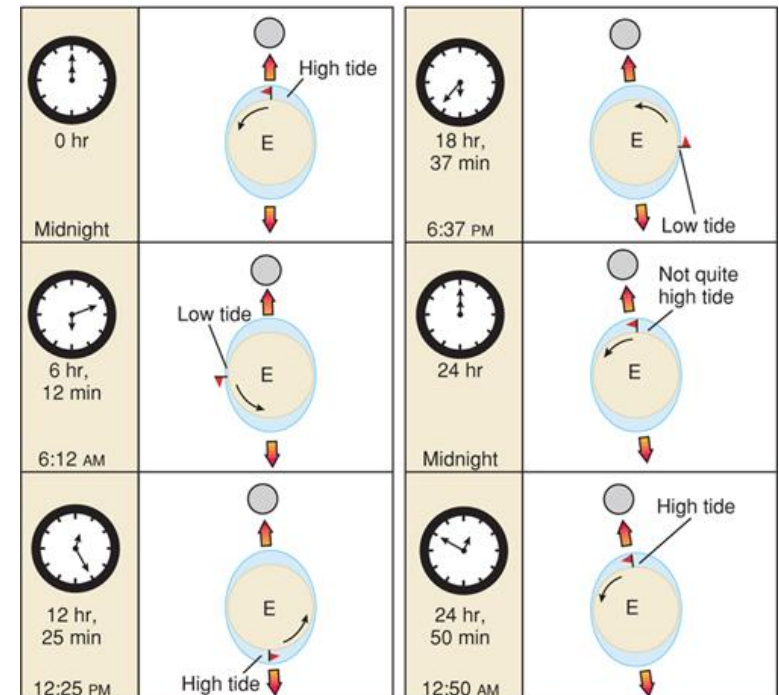
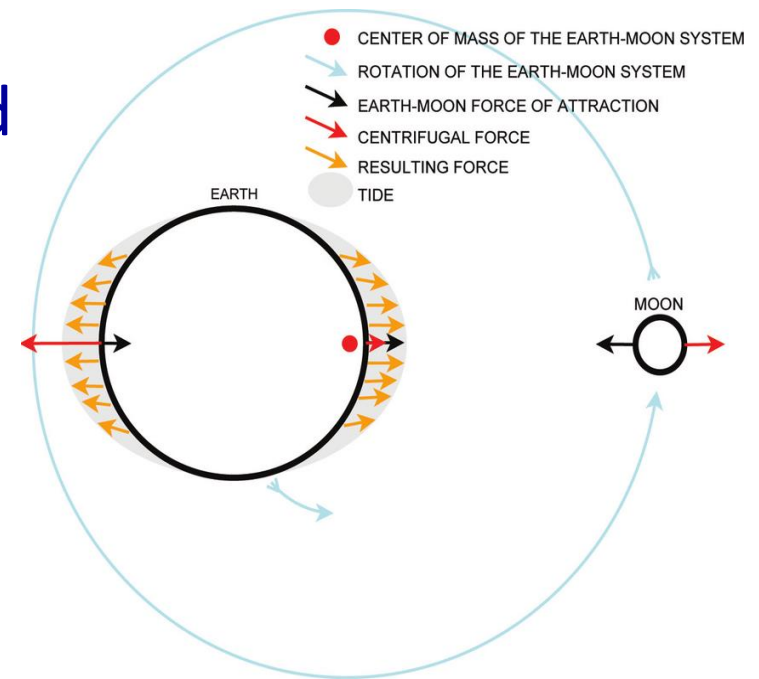


- **Tides** are the rhythmic rising and falling of sea surface levels

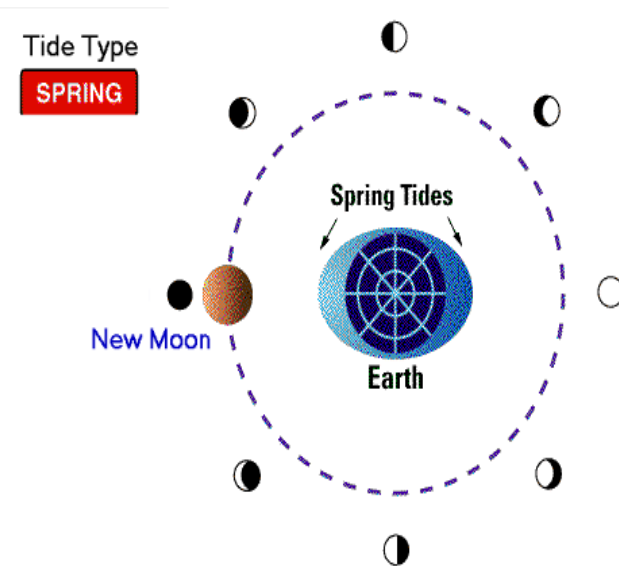
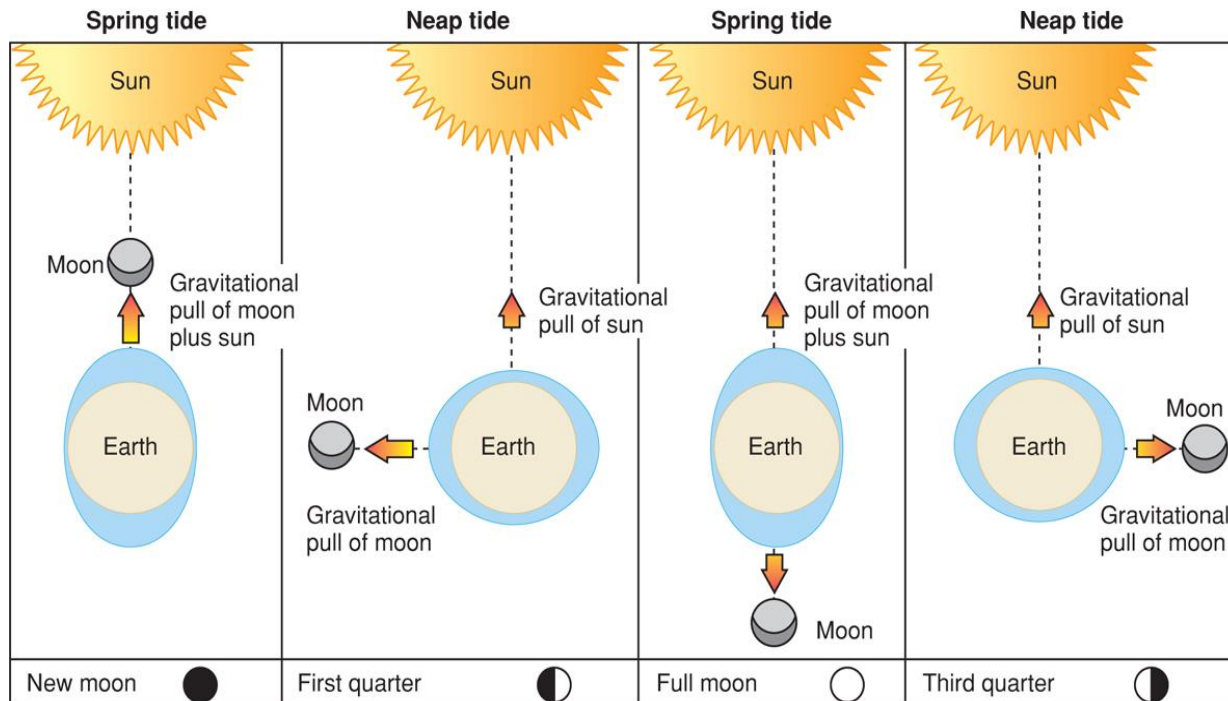
- ❑ The tides are caused by the gravitational pull of the moon on the ocean waters (and the sun to a lesser degree)

- ❑ Waters on the side of the earth closer (or farthest) to the moon are pulled towards (or against) the moon, causing a high tide

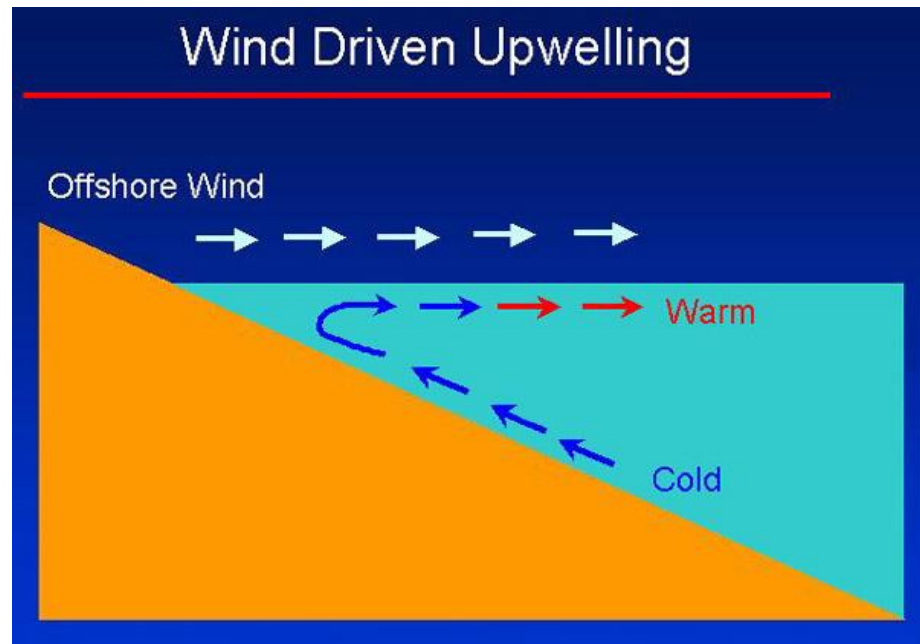
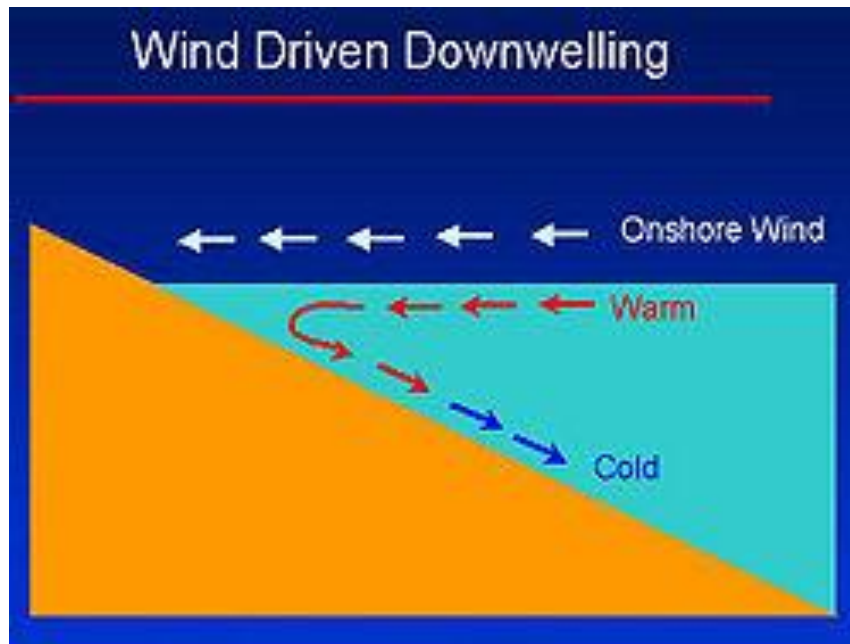
- ❑ Waters on the side of the earth which has angle with moon are pushed away, causing a low tide



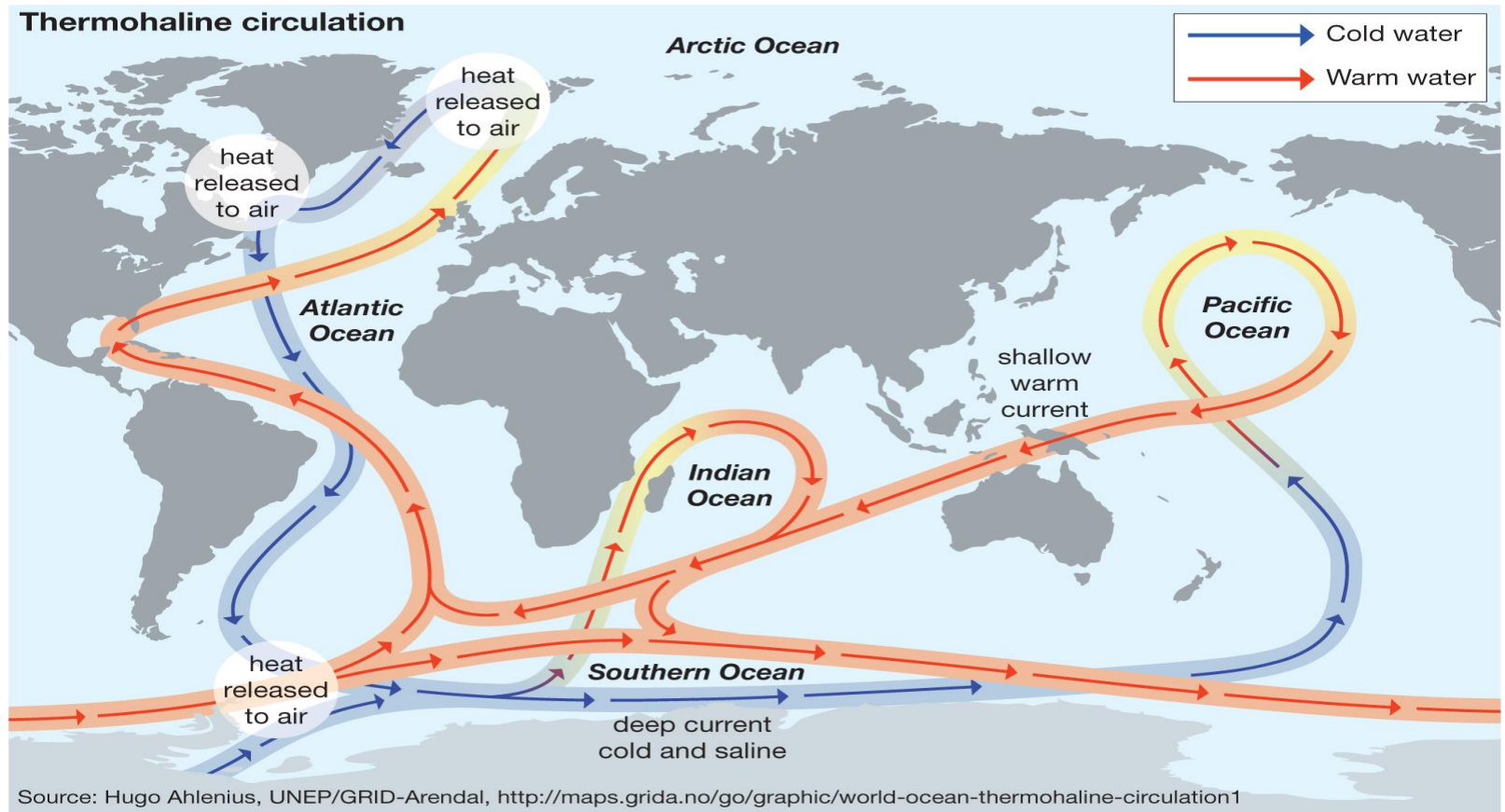
- **Spring tides (涌潮/大潮)** happen each month when the sun and the moon are both in line at times of the new and full moon, the tidal range is even greater
- **Neap tides (小潮)** occur between the spring tides when the sun and moon are not in alignment



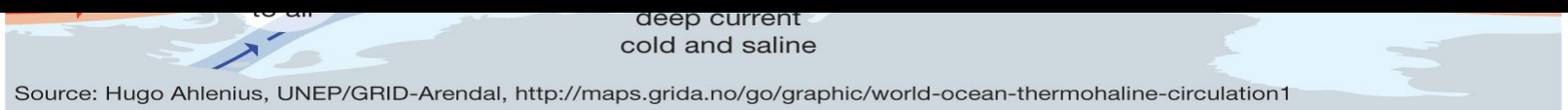
- In some locations, large volumes of water may sink or rise:
 - ❑ Water sinks due to changes in temperature and salinity – this is known as an area of **downwelling** which brings gases from the surface to deeper layers
 - ❑ Areas of **upwelling** come from currents that push deeper waters toward the surface. Nutrients are much more plentiful in the deeper layers, so these areas of upwelling are beneficial for organisms in an upwelling area

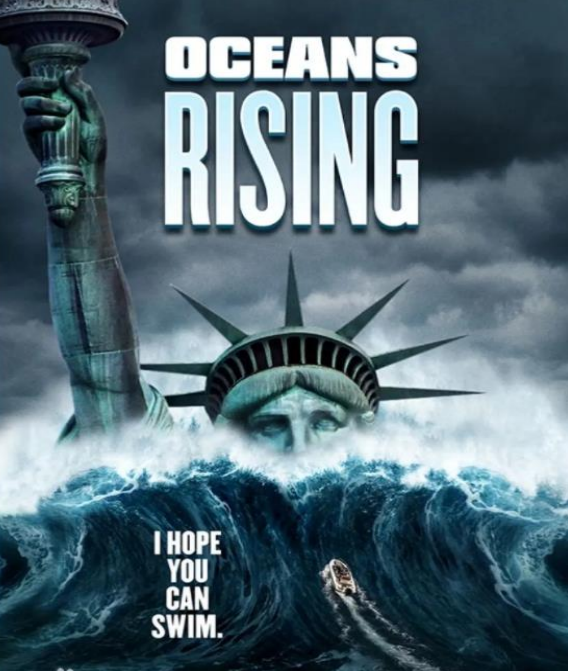


- The great ocean conveyor (大洋输送带)
- ❑ Driven by changes in density, which in turn is determined by temperature and salinity, also called thermohaline circulation
- ❑ Critical in regulating Earth's climate changes, nutrition transportation, gas mixing, *etc.*



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Human activities



Global warming



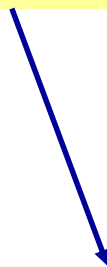
Ice melting



Sea level rising



Submerged



Dilution



Decreased density



Great ocean conveyor



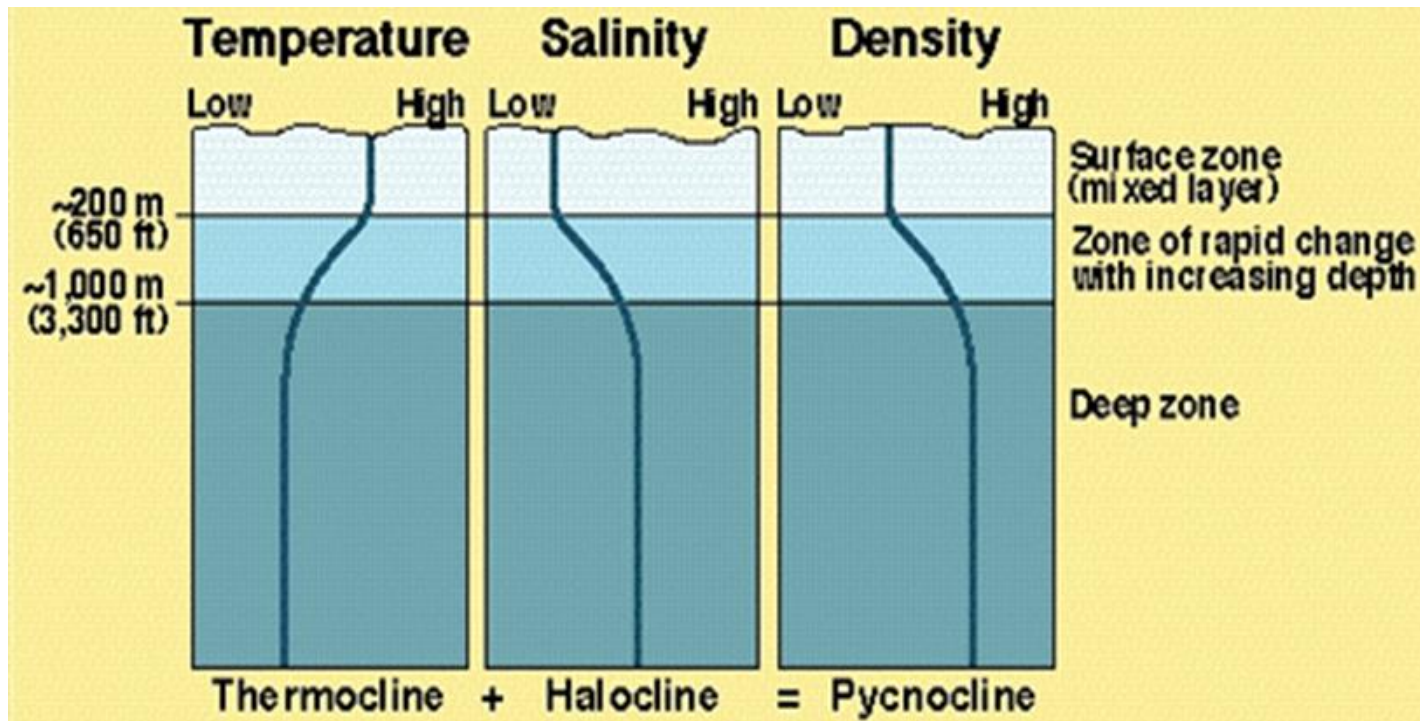
Ice age



Perish of human world



- Despite repeated mixing by winds and currents at the ocean's surface, the ocean is still stratified into three layers:
 - ❑ **Surface layer** – from surface to about 200 meters; this layer stays well mixed most of the year
 - ❑ **Intermediate layer**—from 200 – 1000 meters; major temperature change (thermocline) is located here; less mixing occurs here
 - ❑ **Bottom layer** – below 1000 meters; low mixing and normally uniformly cold



Review of Chapter 3:

1. Water/Seawater and the unique prosperities

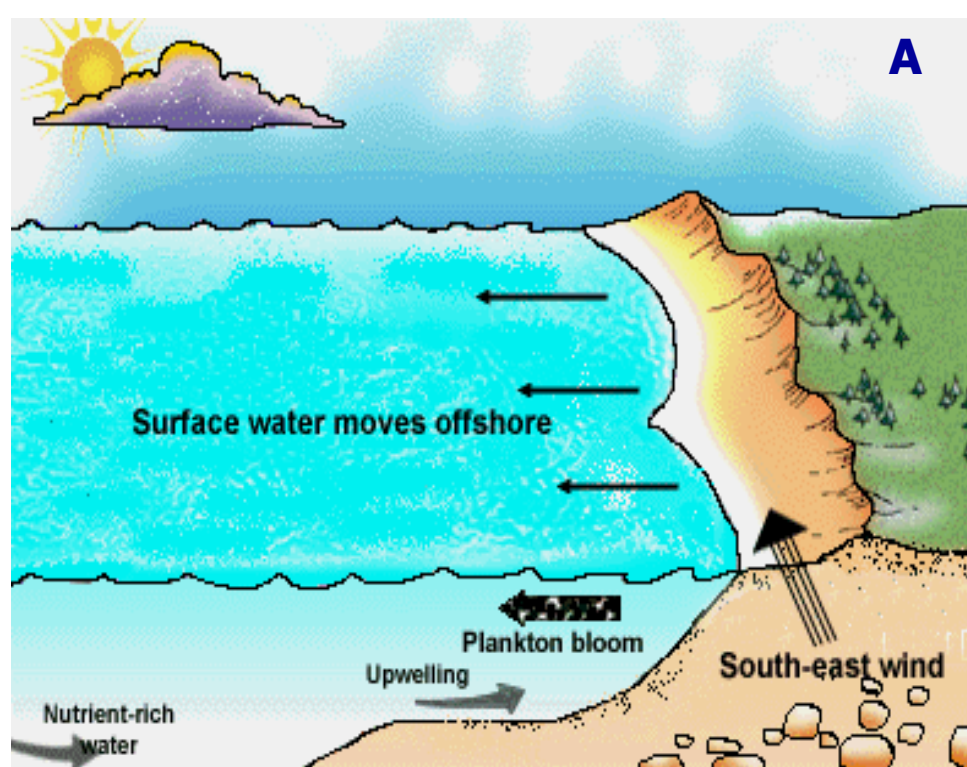
Heat / solvent / seawater composition / temperature / salinity / density / pressures / transparency

2. Ocean circulation

waves, tides, currents and gyres, layers (downwelling and upwelling)

Homework (in English or Chinese):

1. Which hemisphere was presented in figure A, explain the reason
2. Explain why the fishery areas in figure B can be the most famous fishing ranch

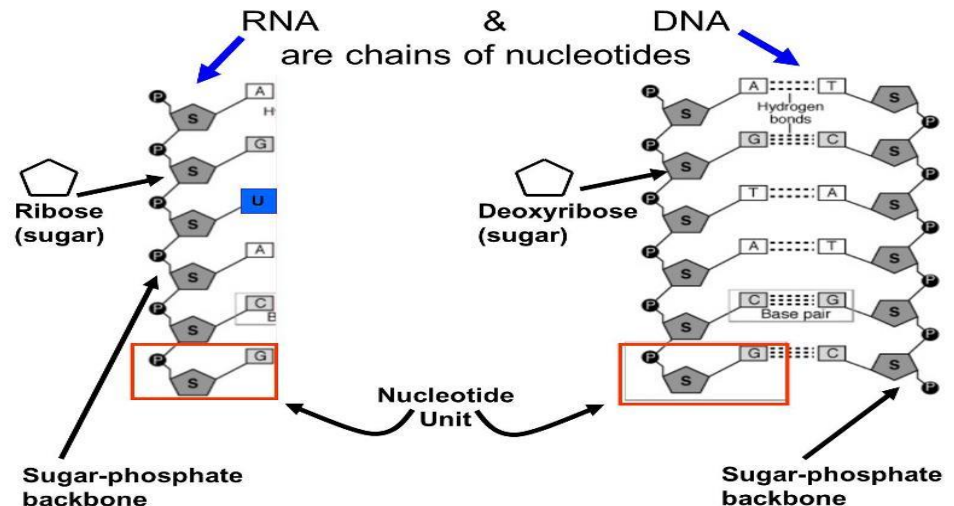




Fundamentals of Biology

Part 1 The ingredients of Life & Living Machinery

- Four organic (contain carbon, hydrogen and oxygen) molecules make up living organisms:
 - Carbohydrates
 - Proteins
 - Lipids
 - Nucleic acids



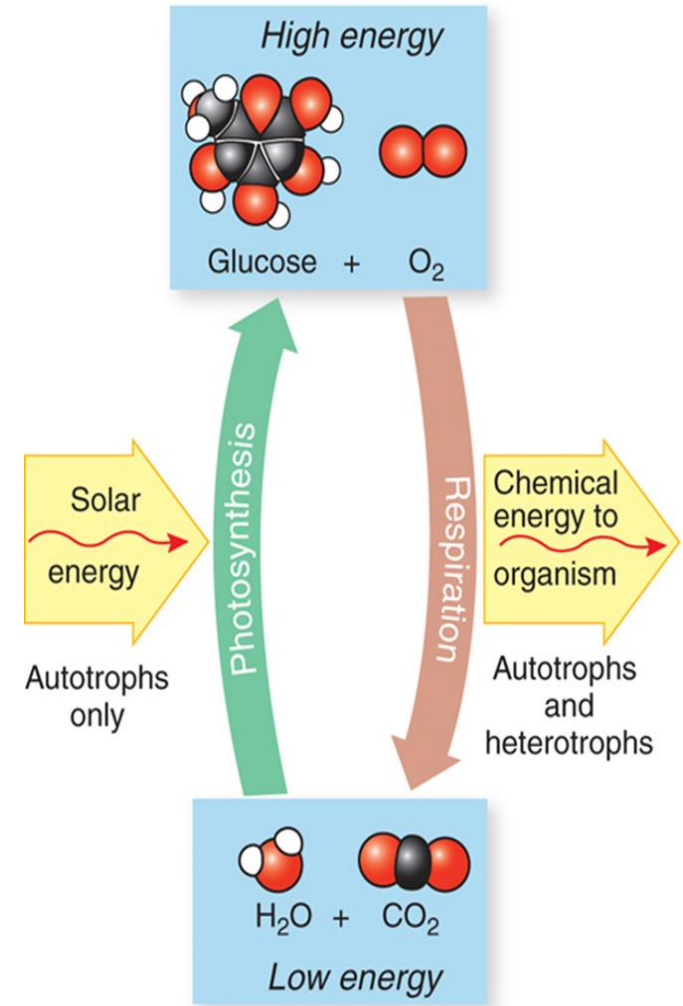
❑ **Photosynthesis:** many organisms use sunlight to produce the energy

❑ **Autotrophs:** organism makes their own carbohydrates

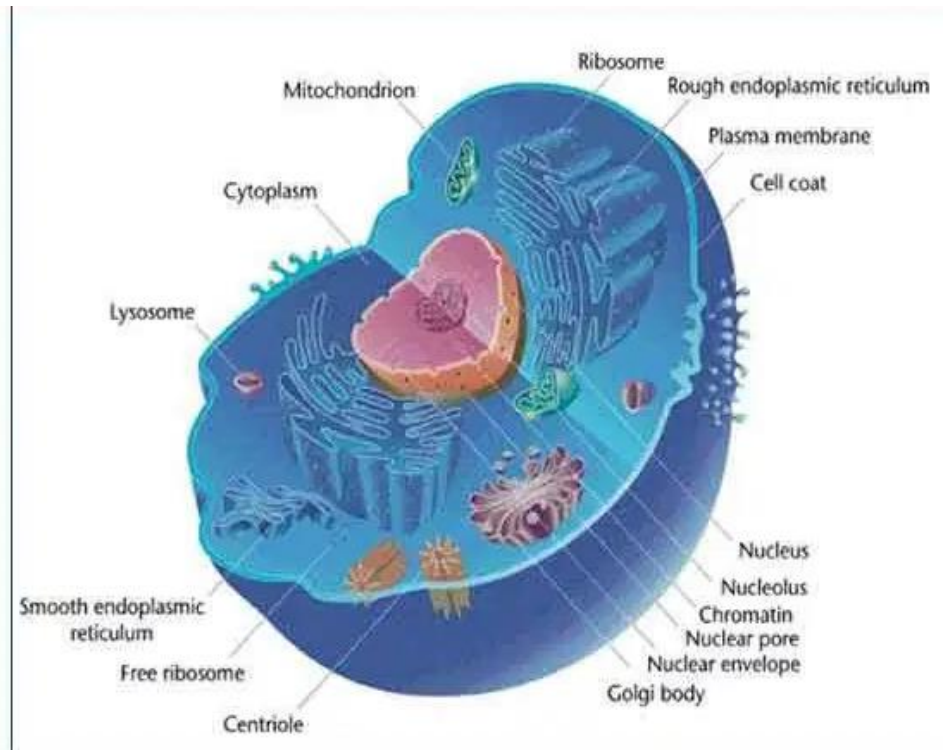
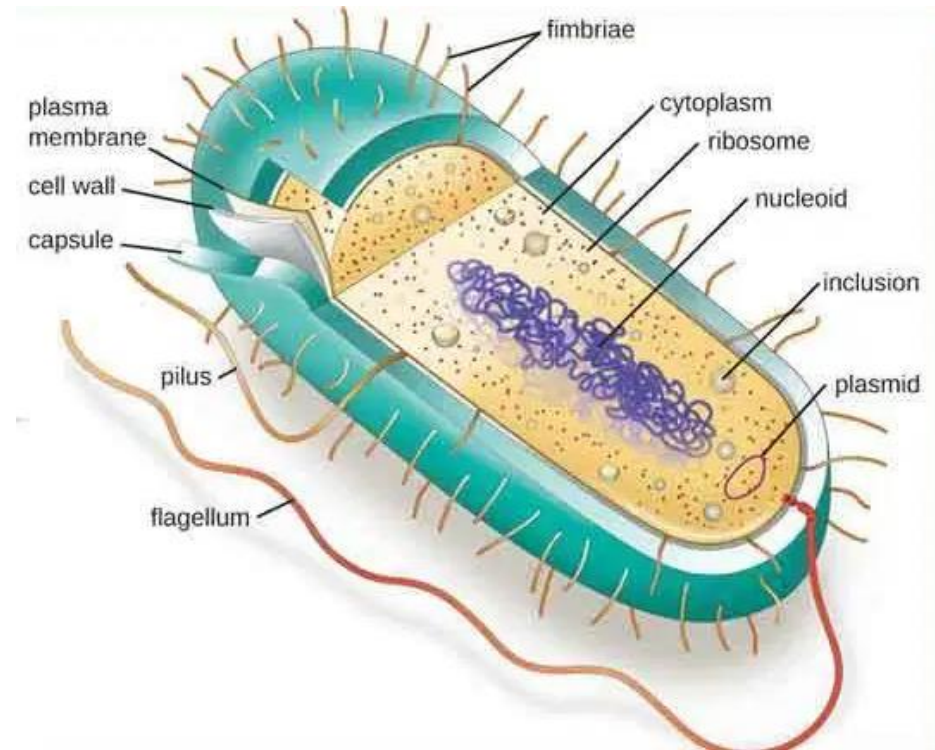
❑ **Heterotrophs:** organism gets carbohydrates by eating other organisms

❑ **Respiration:** the process which organisms must break down the carbohydrates within their cells for energy

❑ Some of the carbohydrates made are converted into other types of molecules such as: Proteins; Lipids; Nucleic acids



- ❑ **Primary production**--the excess produced energy than the autotrophs can use
- ❑ **primary producers**--Organisms responsible for this primary production
- ❑ **Marine organisms are a major source of worldwide primary production**
- ❑ Marine organisms require nutrients to convert carbohydrates to other types of molecules, including minerals, vitamins and even raw elements
- ❑ All living organisms can be divided into two basic groups based on cellular composition:
 1. Prokaryotic
 2. Eukaryotic



- If you forgot the cell structure, you can play the following puzzle game:

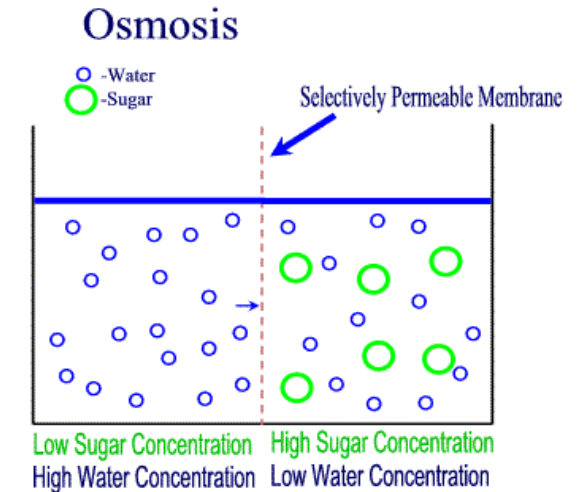
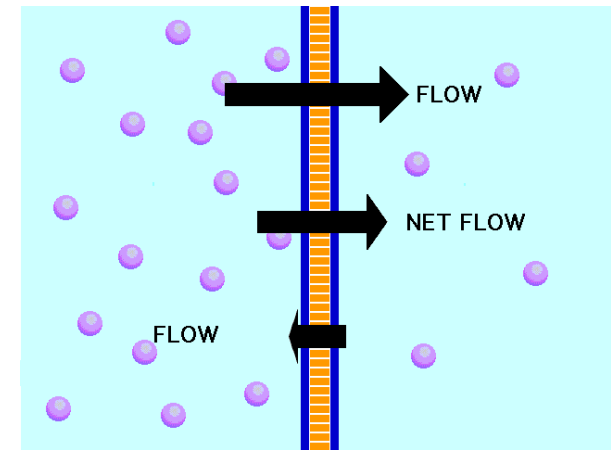
Go to www.cellsalive.com/

- In the middle column labeled “Cell puzzles and games,” click “match, jigsaw & crossword”
- On the “Cells Alive! Puzzle Page,” complete **“Animal Cell”** Jigsaw puzzles

Part 2 Challenges of life in the sea

1. Diffusion and Osmosis

- **Diffusion** (扩散) : **Solutes** will move from areas where they are more concentrated to an area where they are less concentrated
- **Osmosis** (渗透) : movement of **water** from an area where it is more concentrated to an area where it is less concentrated through a semipermeable membrane (半透膜)
- Since marine organisms live in a very solute-rich environment, they have a tendency to gain solutes and lose water
- Resulting the death of cells if the water loss/solute gain is significant
- Organisms must find ways to deal with this diffusion and osmosis



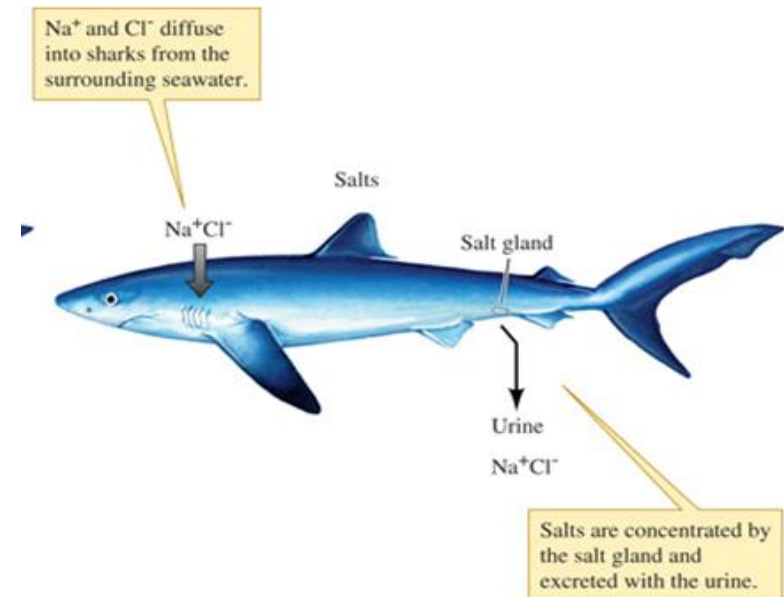
- Method 1—Osmoconformers (变渗生物/渗透压适应者)

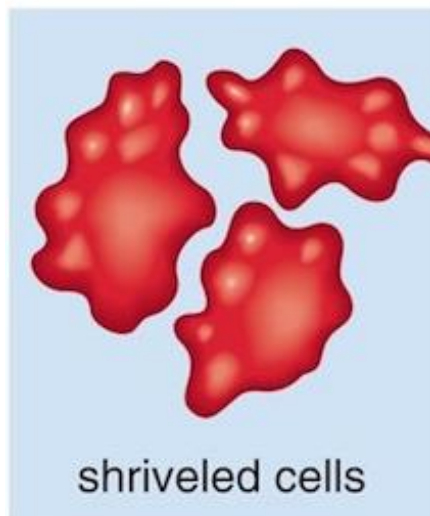
- Do not attempt to control solute/water balance
- Their internal concentration varies as the salinity in the water around them changes: many invertebrate, e.g. crabs, clams, sea squirts
- Most can only tolerate a very narrow range of salinity



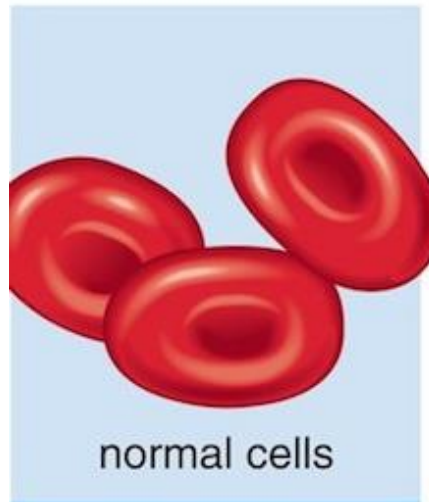
- Method 2—Osmoregulators (渗透压调节者)

- Organisms control their internal concentrations
- Can generally tolerate a wider range of salinities than osmoconformers
- Osmoregulation can be done in a variety of ways such as secreting very little urine or using specialized glands to secrete salts as examples

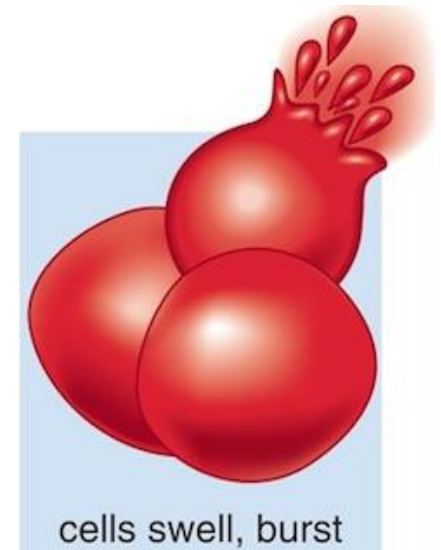




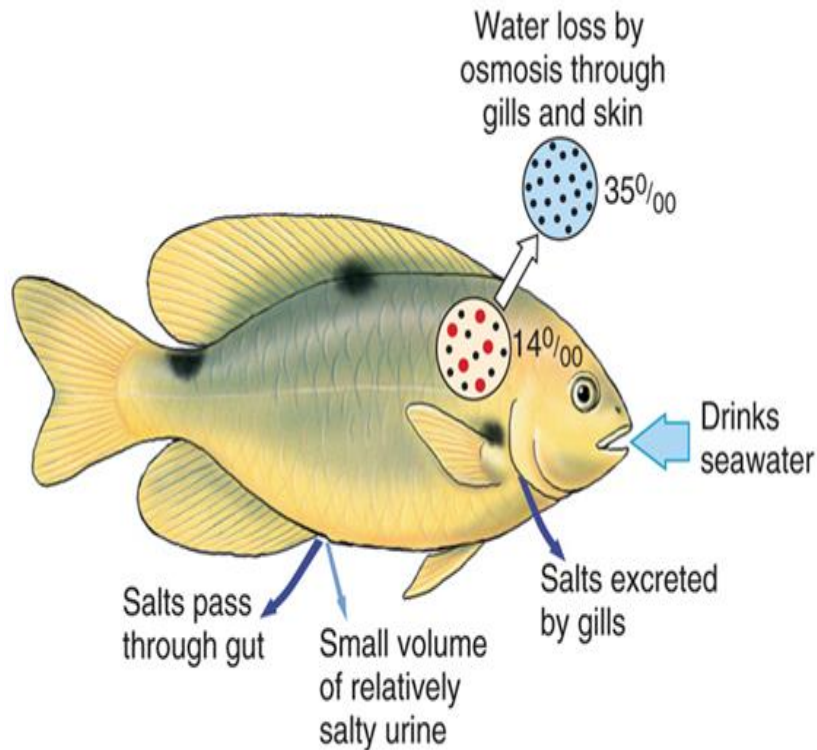
shriveled cells



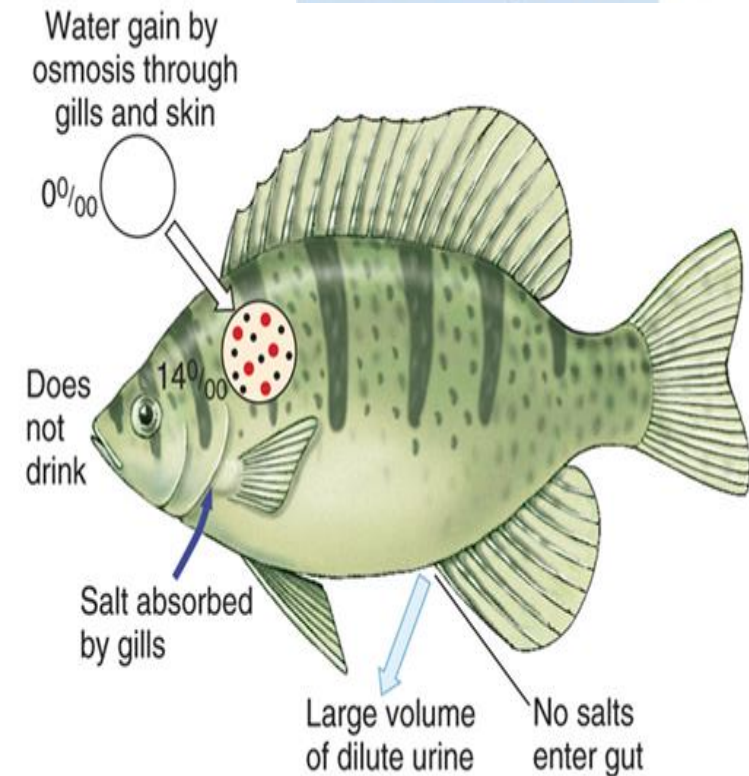
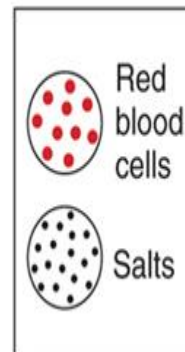
normal cells



cells swell, burst



(a) Marine fish



(b) Freshwater fish

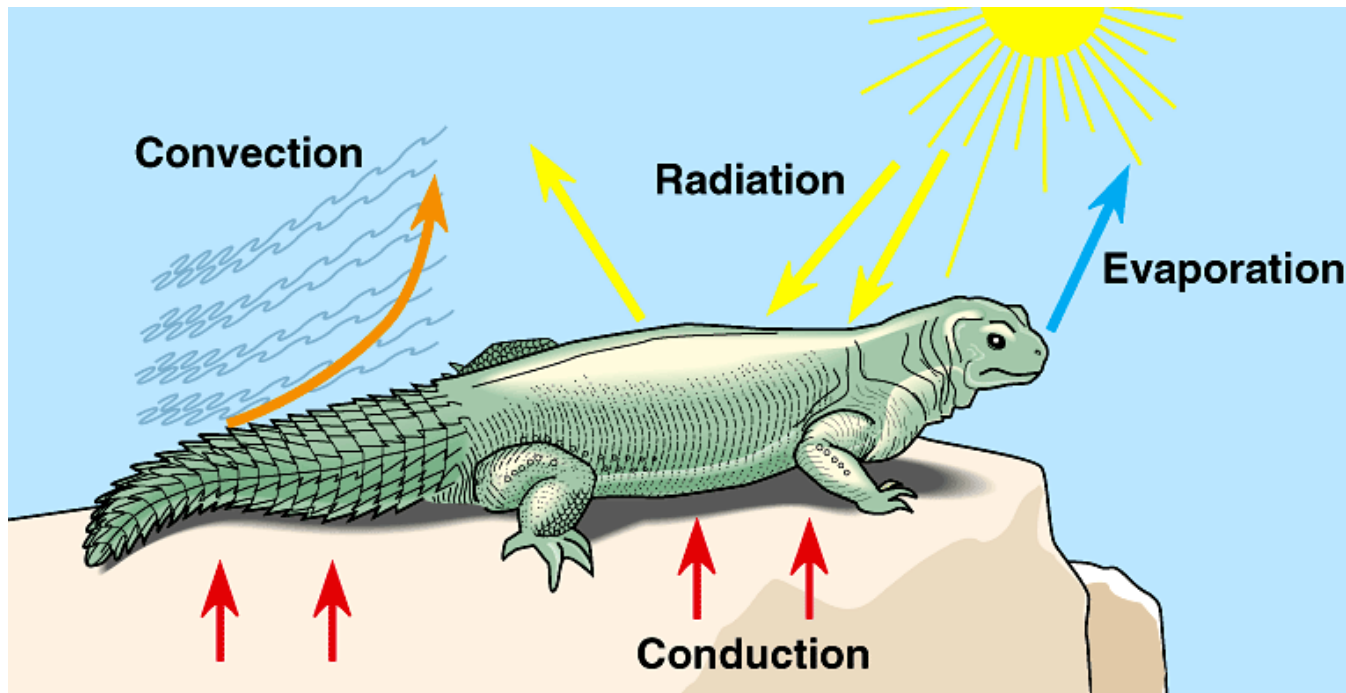
2. Temperature Control

- Categorized according to how their metabolism affects their body temperature

1) Ectotherms (外温动物/变温动物/冷血动物)

- Generate body heat metabolically, the heat is quickly lost to environment, internal body temperature dose not raise

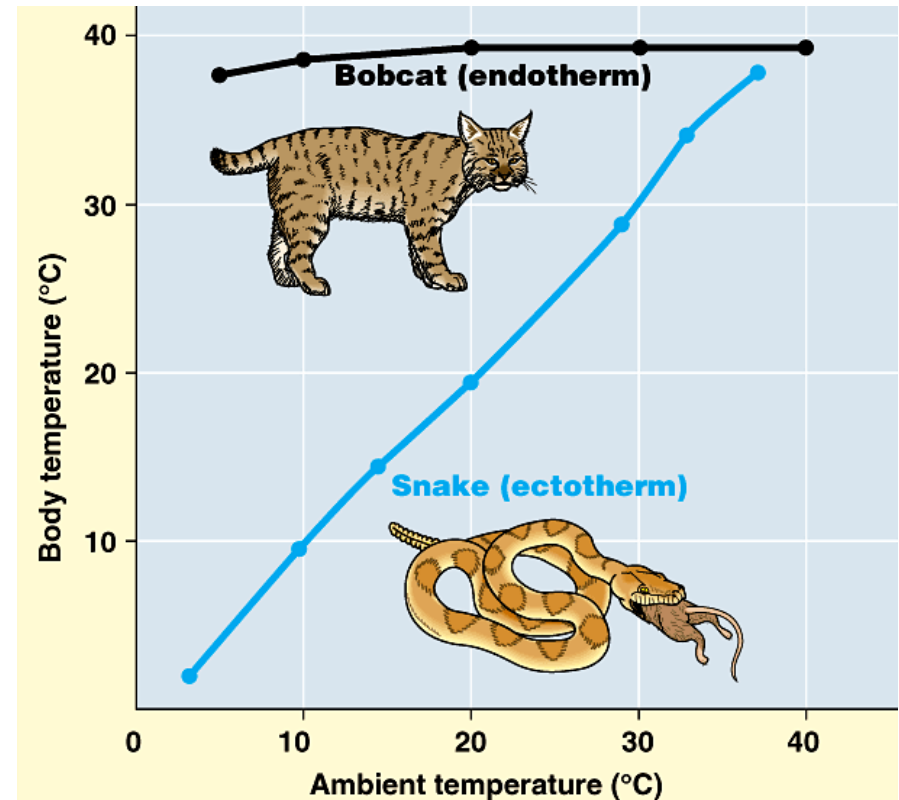
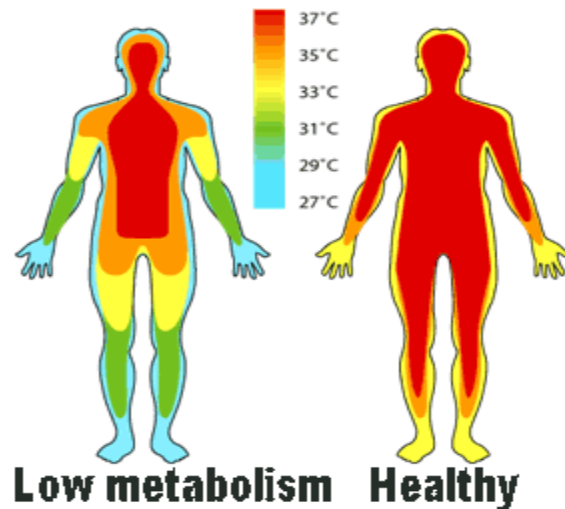
Examples: plants, all invertebrates, fish, amphibians etc.



2) Endotherms (内温动物/恒温动物)

- Generate body heat metabolically and retained, body temperature raised above that of the surrounding environment

All birds and mammals, some large shark, tuna



- Categorized according to whether organisms can keep the body temperature more or less constant

1) Poikilotherms (变温动物)

- Body temperature mimics the surrounding environment. They do not regulate their metabolisms to heat or cool themselves
- Ectotherms are poikilotherms

2) Homeotherms (恒温动物)

- These organisms retain metabolic heat and can control metabolism to maintain a constant internal temperature
- Homeotherms are endotherms

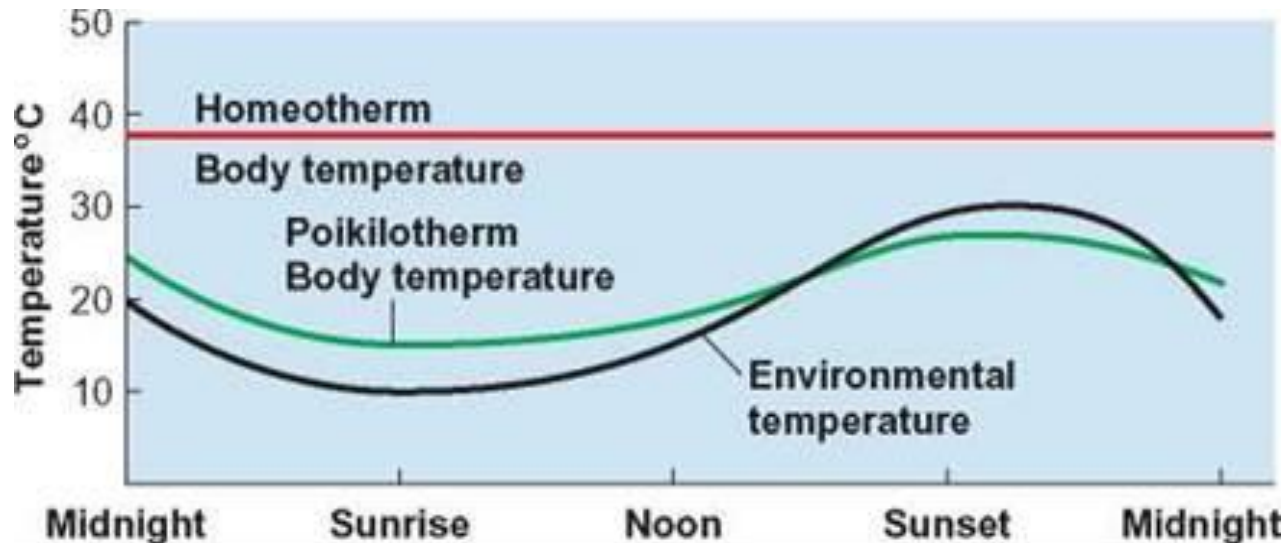


Table 4.2 Classification of Animals by Their Response to Changes in External Temperature

Poikilotherms:

Temperature varies with external temperature

Homeotherms:

Regulate body temperature so it doesn't vary as much as external temperature

Ectotherms: Most metabolic heat is rapidly lost to the environment; body temperature matches external temperature

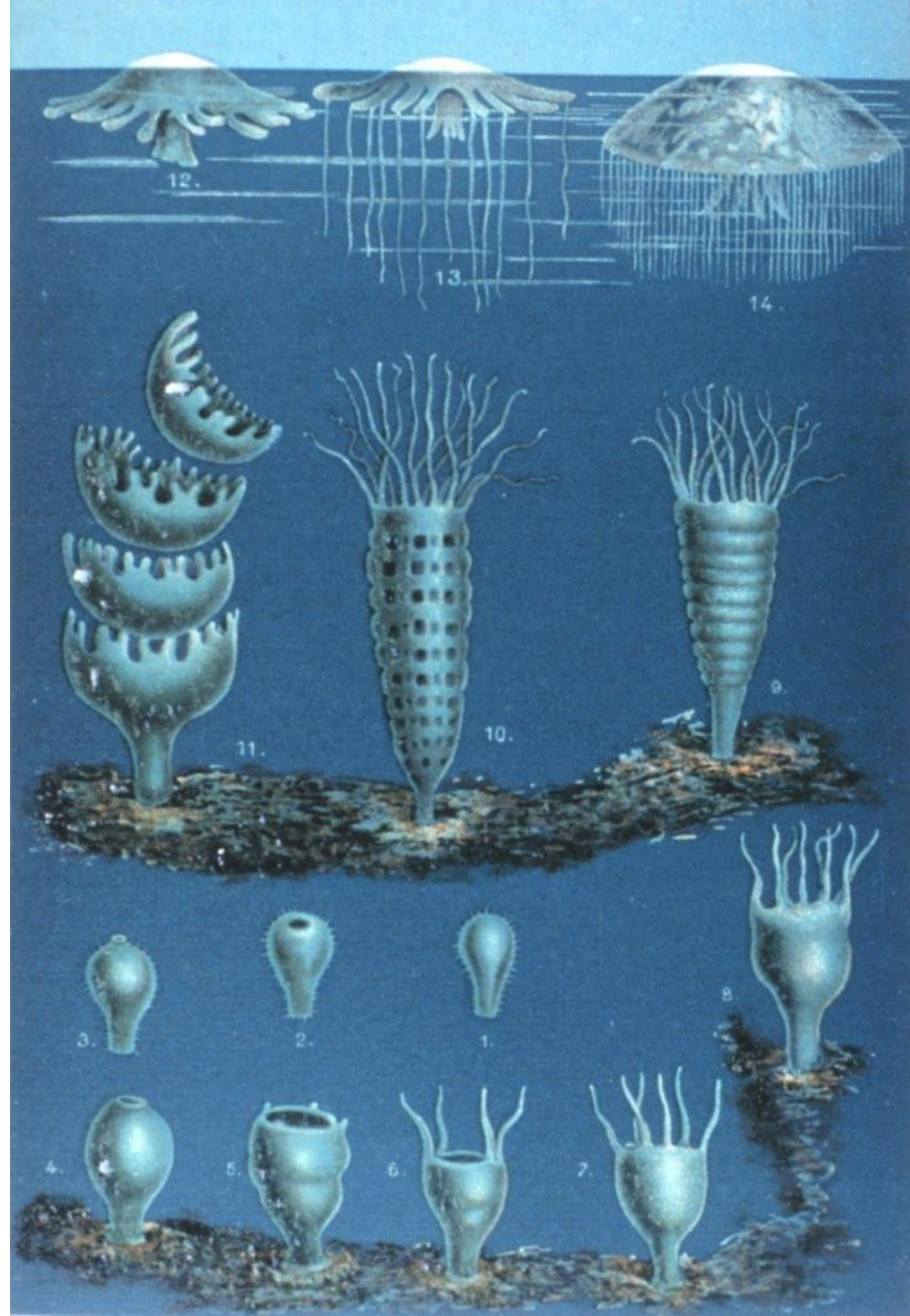
Endotherms:

Retain significant metabolic heat; internal temperature stays warmer than external temperature

Part 3 Reproduction

1. Asexual reproduction

- Does not involve mating of two individuals
- Young are produced by a single parent organism
- The young produced are genetically identical to the parent



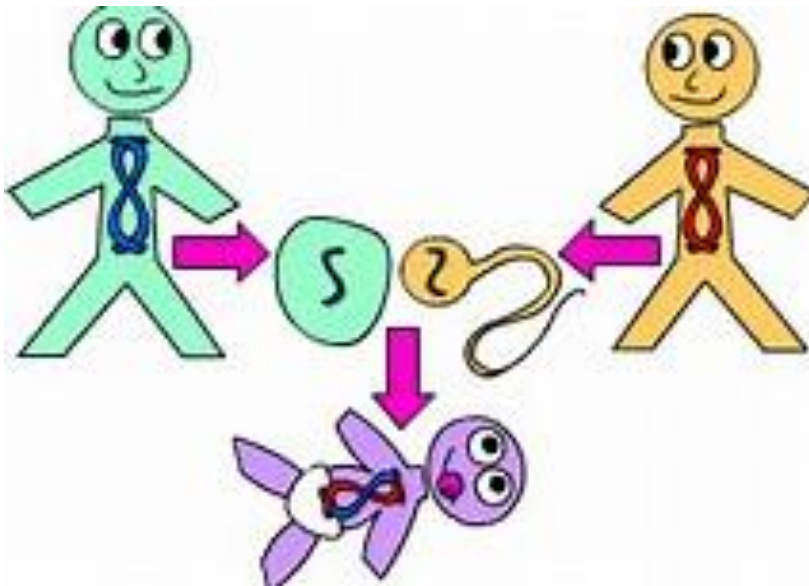
- Examples of Asexual Reproduction

- **Fission:** the splitting of one organism into two smaller organisms of equal size
- **Budding:** the organism develops buds (small clones) that eventually break off and become another organism
- **Vegetative reproduction:** a plant reproduces new individuals by sending an underground stem (rhizome) sideways from which new plants will sprout



2. Sexual reproduction

- Normally involves two individuals
- Parent individuals produce gametes (eggs or sperm) that unite to produce a new, genetically unique individual
- Ovaries are the organs that produce eggs
- Testes are the organs that produce sperm

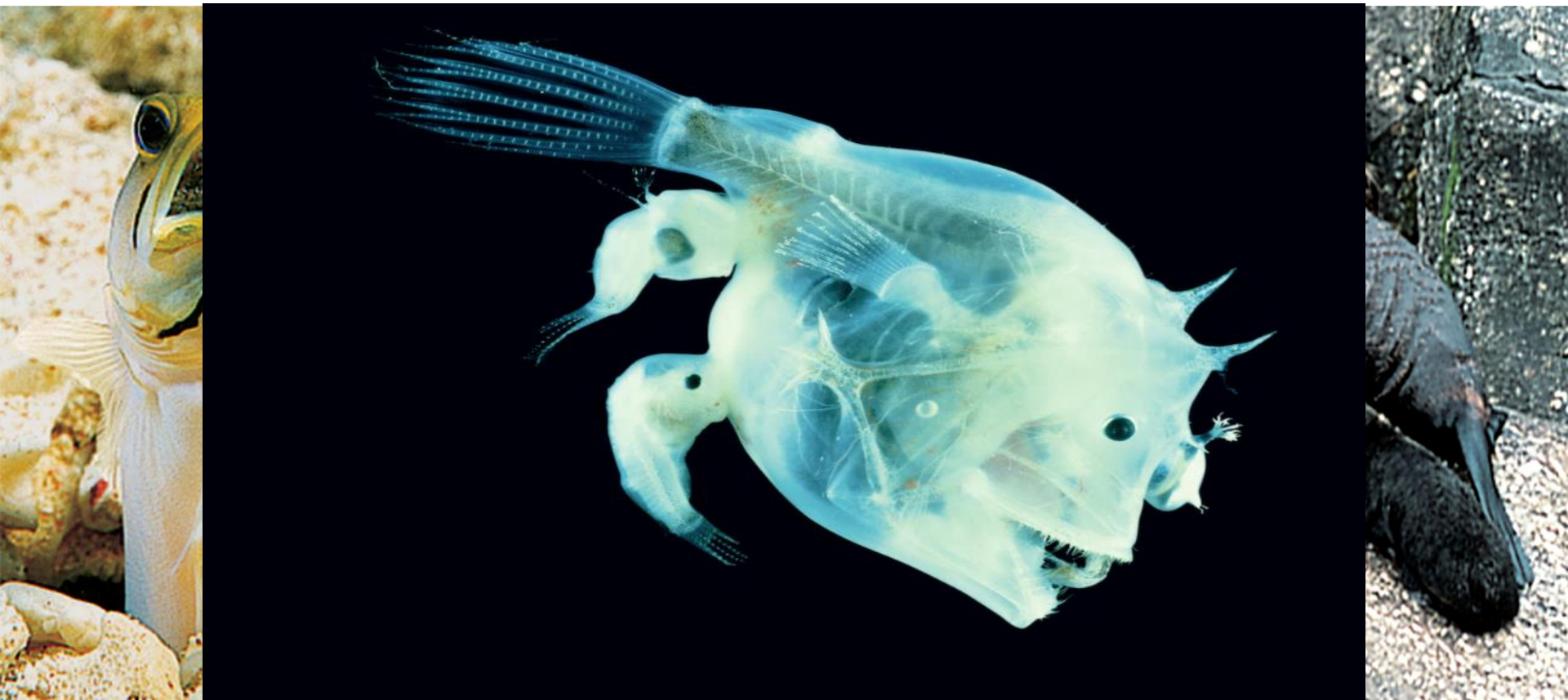


3. Strategy of reproduction:

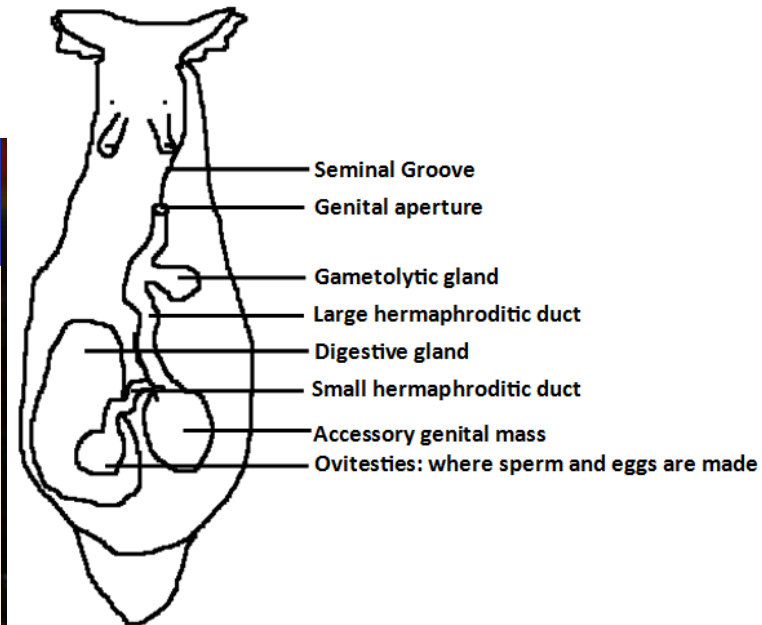
- broadcast spawning (散播式产卵): Many marine organisms release their eggs and sperm directly into the water
 - For broadcast spawning to be effective, millions of gametes must be released into the water at roughly the same time to ensure fertilization will occur
 - Many broadcast spawning species time the release of their eggs to tides, moon phase, water temperature, etc. to ensure success



- Other marine organisms rely on internal fertilization, where a copulatory organ is used to insert sperm directly into the female's reproductive tract; this method requires contact between parent individuals, but less gametes are required for success



- Hermaphrodites (雌雄同体) – individuals that have male and female reproductive tissues either simultaneously or at different phases during the life, Examples:
 - Protandry (雄蕊先熟) : an individual spends the first portion of the life as a functional male then becomes a female later in life after some cue initiates the change
 - Protogyny (雌蕊先熟): an individual spends the first portion of the life as a functional female then becomes a male later in life after some cue initiates the change



Homework:

Find and read the literature, answer the following questions (in English):

1. How whale sharks retain their body temperature?
2. Which thermoregulation strategy it can belong to?
3. What are the main research methods?

RESEARCH ARTICLE

Body temperature stability in the whale shark, the world's largest fish

Itsumi Nakamura^{1,*}, Rui Matsumoto² and Katsufumi Sato³